Junchuan Zhao

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EDUCATION

National University of Singapore (NUS) Jan. 2024 – Dec. 2027 (Expected)
- **Ph.D.** *in Computer Science*
  - *Advisor: Prof. Ye Wang, Sound and Music Computing (SMC) Lab*

National University of Singapore (NUS) Aug. 2022 – Dec. 2023
- **M.Sc.** *in Computer Science, specialization in AI* GPA: *4.75/5.00*
  - *Advisor: Prof. Ye Wang*

Beijing University of Posts and Telecommunications (BUPT) Sep. 2018 – Jun. 2022
- **B.Sc.** *in Telecommunication Engineering with Management* GPA: *91.44/100*
  - *Professional Ranking: 4/319*

Queen Mary University of London (QMUL) Sep. 2018 – Jun. 2022
- **B.Sc.** *in Telecommunication Engineering with Management (Dual Degree Program)*
  - *First Class Degree*

RESEARCH INTERESTS

Speech Generation · Controllable & Expressive Speech Synthesis · Speech Language Models · Instruction-Following TTS · Singing Voice Synthesis · Music Generation & Analysis

PUBLICATIONS

* indicates equal contribution; † indicates corresponding author

[1] **J. Zhao**, M. D. Vu, and Y. Wang. "MSpoof-TTS: Multi-Resolution Spoof-Guided Inference for Discrete Speech Synthesis." *The 27th Annual Conference of the International Speech Communication Association* (**Interspeech**), 2026. arXiv:2603.05373

[2] Q. Liang*, Y. Liu*, R. Wei*, N. Lu, **J. Zhao**†, and Y. Wang. "TED-TTS: Training-Free Intra-Utterance Emotion and Duration Control for Text-to-Speech Synthesis." *The 64th Annual Meeting of the Association for Computational Linguistics* (**ACL Main**), 2026. arXiv:2601.03170

[3] **J. Zhao**, W. Zeng, T. Lyu, and Y. Wang. "CoMelSinger: Discrete Token-Based Zero-Shot Singing Synthesis With Structured Melody Control and Guidance." *IEEE Transactions on Audio, Speech and Language Processing* (**TASLP**), 2026. DOI: 10.1109/TASLPRO.2026.3664643

[4] W. Zeng, **J. Zhao**, and Y. Wang. "Bridging Piano Transcription and Rendering via Disentangled Score Content and Style." *The Fourteenth International Conference on Learning Representations* (**ICLR**), 2026. arXiv:2509.23878

[5] T. Lyu*, **J. Zhao***, and Y. Wang. "KSDiff: Keyframe-Augmented Speech-Aware Dual-Path Diffusion for Facial Animation." *IEEE International Conference on Acoustics, Speech, and Signal Processing* (**ICASSP**), 2026. arXiv:2509.20128

[6] Z. Li, **J. Zhao**†, F. Lee, and A. Yee. "InconVAD: A Two-Stage Dual-Tower Framework for Multimodal Emotion Inconsistency Detection." *IEEE International Conference on Acoustics, Speech, and Signal Processing* (**ICASSP**), 2026. arXiv:2509.20140

[7] **J. Zhao***, X. Wang*, and Y. Wang. "Prosody-Adaptable Audio Codecs for Zero-Shot Voice Conversion via In-Context Learning." *The 26th Annual Conference of the International Speech Communication Association* (**Interspeech**), 2025. DOI: 10.21437/Interspeech.2025-464

[8] **J. Zhao**, L. Chetwin, and Y. Wang. "SPSinger: Multi-Singer Singing Voice Synthesis with Short Reference Prompt." *IEEE International Conference on Acoustics, Speech, and Signal Processing* (**ICASSP**), 2025. DOI: 10.1109/ICASSP49660.2025.10888907

[9] **J. Zhao**, L. Chetwin, and Y. Wang. "SinTechSVS: A Singing Technique Controllable Singing Voice Synthesis System." *IEEE/ACM Transactions on Audio, Speech and Language Processing* (**TASLP**), 2024. DOI: 10.1109/TASLP.2024.3394769

PREPRINTS

[1] M. D. Vu*, **J. Zhao***, and Y. Wang. "Does Expert Supervision Improve Text-to-Speech? An Empirical Study of Preference Optimization." 2026. arXiv:2610.25292

[2] X. Wang, **J. Zhao**†, and Y. Wang. "X-MDD: LLM-Based Mispronunciation Detection and Diagnosis for Unseen Languages with Canonical References." 2026. arXiv:2610.23079

[3] **J. Zhao**, C. Xu, W. Zeng, H. Li, Y. Guo, and Y. Wang. "EDICT: Edit Who Speaks, Control How They Speak – Global Timbre Editing and Local Instruction Control for TTS." 2026. arXiv:2610.11437

[4] X. Wang, **J. Zhao**, and Y. Wang. "Joycent: Diffusion-based Accent TTS without Accented Phone Prediction." 2026. arXiv:2606.16417

[5] H. Li, C. Xu, **J. Zhao**, Y. Cao, L. Xue, Y. Guo, and E. S. Chng. "Preference Optimization for Non-Verbal Vocalization Synthesis." 2026. arXiv:2608.24163

[6] **J. Zhao**, M. D. Vu, B. Zhang, and Y. Wang. "AnyBand: Unified Multi-Bandwidth Speech Extension via Frequency-Aware In-Context Spectral Infilling." 2026. arXiv:2608.00572

[7] **J. Zhao***, Q. Liang*, and Y. Wang. "PersonaGest: Personalized Co-Speech Gesture Generation with Semantic-Guided Hierarchical Motion Representation." 2026. arXiv:2605.07252v1

[8] L. Chen, **J. Zhao**†, and Y. Wang. "Remix the Timbre: Diffusion-Based Style Transfer Across Polyphonic Stems." 2026. arXiv:2605.09259

[9] B. Zhang*, **J. Zhao***, A. Madhukumar, Y. Wang, and I. McLoughlin. "CodecFlow: Efficient Bandwidth Extension via Conditional Flow Matching in Neural Codec Latent Space." 2026. arXiv:2603.02022

INTERNSHIP EXPERIENCE

Research Intern, Tencent July 2026 – Now
Mentor: Speech Synthesis Research Team, Tencent Singapore

- Explored prompt-based timbre and style adaptation techniques to enable flexible voice customization and fine-grained speech style control from reference speech.
- Proposed **EDICT** (Demo, arXiv:2610.11437), a framework unifying global timbre editing and local expressive control for instruction-based TTS by generating an edited acoustic reference in codec-token space that anchors speaker identity for a frozen backbone, with KV-cache rebuilding at segment boundaries to balance instruction adherence, speaker consistency, and transition quality.

Research Intern, Tencent May 2026 – July 2026
Mentor: Speech Synthesis Research Team, Tencent Singapore

- Conducted research on instruction-following text-to-speech systems, focusing on controllable speech generation from natural language instructions.
- Investigated multi-instruction TTS frameworks for simultaneous control of speaking style, emotion, prosody, and duration within a single utterance.
- Proposed **EmoSimplex** (Demo), a training-free inference-time framework that organizes an instruction-based autoregressive TTS model into a continuous, compositional emotion control space via a bank of ordered instruction anchors and a unified affine operator over next-token logits, enabling intensity control, mixed-emotion synthesis, and text-emotion suppression without backbone modification or activation access.

Research Assistant, Tsinghua University, NLP Lab Jun. 2023 – Oct. 2023
Mentor: Dr. Yuan Yao, Tsinghua University

- Conducted a comprehensive literature review on multimodal large language models (audio $\leftrightarrow$ text), generalized audio understanding, and neural audio synthesis.
- Reviewed and categorized representative works on audio LLMs and audio-text models to guide architectural design.
- Explored training a unified model architecture for joint text and audio modeling, with a focus on improving cross-modal alignment between textual and acoustic representations.

MISCELLANEOUS

Languages	Mandarin (native), English (professional proficiency)	
Skills	Python, Java, JavaScript, MATLAB	
Services	Reviewer for TASLP, ToMM, ICASSP, Interspeech, ACM MM, ISMIR, EMNLP, SLT, AAAI, NeurIPS	
	Event chair and performer at NUS SMC Concert	2022 – 2025
	Jazz Vocalist at NUS Jazz Band	2022 – 2026
Teaching	CS3244 Machine Learning	AY2025/26
	CS5647 Sound and Music Computing	AY2024/25, AY2025/26
Awards	SoC Honour List of Student Tutors	AY2024/25
	SoC's Teaching Fellowship Scheme (TFS)	2025
	NUS Research Scholarship	2023 – 2027
	Queen Mary University of London Undergraduate College Prize (14/600)	2022
	Outstanding College Student in Beijing	2022